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4 / 4 submitted

Level 2 — 25 marks/question

L2-Q1. (M) Max-Min

J

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0;i<n;i++){
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        int M = sc.nextInt();
        int N = sc.nextInt();

        if((M > 0 && N > 0) && (M<=n && N<=n)){
            System.out.printf("%dth Maximum Number = %d\n",M,arr[n-M]);
            System.out.printf("%dth Minimum Number = %d\n",N,arr[N-1]);
            System.out.printf("Sum = %d\n", (arr[n-M]+arr[N-1]));
            System.out.printf("Difference = %d", (arr[n-M]-arr[N-1]));
        }else{
            System.out.println("M and N values are Invalid.");
        }
    }
}
```

INPUT

7
14
16
87
36
25
89
34
1
3

OUTPUT

1th Maximum Number = 89
3th Minimum Number = 25
Sum = 114
Difference = 64

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L2-Q2. (M) Perfect Num. Gen.

J

CODE

```
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int k = 0;

        if(n<0){
            System.out.println("Given number is negative.");
        }else if(n==0){
            System.out.println("Given number is zero.");
        }else{
            int[] arr = new int[n];
            for(int i = 1;i<10000;i++){
                int sum = 0;
                for(int j = 1;j<i;j++){
                    if(i%j==0){
                        sum += j;
                    }
                }
                if(sum == i){
                    k = k + 1;
                    if(k==n){
                        arr[k-1] = i;
                    }
                }
            }

            System.out.printf("First %d perfect numbers are: ",n);

            for(int i = 0;i<n;i++){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

INPUT

3

INPUT	3
OUTPUT	First 3 perfect numbers are: 6 28 496

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L2-Q3. (M) Student

j	
CODE	<pre> import java.util.*; class Main { public static void main(String[] args){ Scanner sc = new Scanner(System.in); double a = sc.nextDouble(); double b = sc.nextDouble(); double c = sc.nextDouble(); double d = sc.nextDouble(); if((a<0 b<0 c<0 d<0) (a>100 b>100 c>100 d>100)){ System.out.println("Marks are invalid."); } else{ double total = a+b+c+d; double avg = (total)/4; String grade; if(avg >= 75){ grade = "Distinction"; } else if (avg >= 60 && avg <75){ grade = "First Division"; } else if(avg >= 50 && avg < 60){ grade = "Second Division"; } else if(avg >= 40 && avg < 50){ grade = "Third Division"; } else{ grade = "Fail"; } System.out.println("Total: "+total); System.out.println("Aggregate: "+avg); System.out.println(grade.toUpperCase()); } } } </pre>
INPUT	90 91 92 93
OUTPUT	Total: 366.0 Aggregate: 91.5 DISTINCTION

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L2-Q4. (M) Palindrome

j	
CODE	<pre> import java.util.*; class Main{ public static void main(String[] args){ Scanner sc = new Scanner(System.in); int ch = sc.nextInt(); if(ch==1){ String s1 = sc.nextLine().trim(); String s2 = sc.nextLine().trim(); </pre>

```
Scanner sc = new Scanner(System.in);
double a = sc.nextDouble();
double b = sc.nextDouble();
double c = sc.nextDouble();
double d = sc.nextDouble();
if((a<0 || b<0 || c<0 || d<0) || (a>100 || b>100 || c>100 || d>100)){
    System.out.println("Marks are invalid.");
}
else{
    double total = a+b+c+d;
    double avg = (total)/4;
    String grade;

    if(avg >= 75){
        grade = "Distinction";
    }
    else if (avg >= 60 && avg <75){
        grade = "First Division";
    }
    else if(avg >= 50 && avg < 60){
        grade = "Second Division";
    }
    else if(avg >= 40 && avg < 50){
        grade = "Third Division";
    }
    else{
        grade = "Fail";
    }

    System.out.println("Total: "+total);
    System.out.println("Aggregate: "+avg);
    System.out.println(grade.toUpperCase());
}
}
```

INPUT	90 91 92 93
OUTPUT	Total: 366.0 Aggregate: 91.5 DISTINCTION

```
j
CODE
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int ch = sc.nextInt();

        if(ch==1){
            String s1 = sc.nextLine().trim();
            String s2 = s1;
            if(s1.equalsIgnoreCase(s2)){
                System.out.println("Palindrome");
            }
            else{
                System.out.println("not a Palindrome");
            }
        }
        else if(ch==2){
            int n = sc.nextInt();
            int m = n;
            int rev = 1;
            while(m != 0){
                m = m % 10;
                rev += rev * m;
                m /= 10;
            }
            if(n == rev){
                System.out.println("Palindrome");
            }
            else{
                System.out.println("Not a Palindrome");
            }
        }
        else{
            System.out.println("Choice is invalid.");
        }
    }
}
```

INPUT	1 Madam
OUTPUT	Palindrome